



Week 3 Internship Report

Network Administration Internship Program – IT-SIMPLERA Institute

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Enterprise Network Design & Configuration

Advanced Routing Configuration using Multi-area OSPF, RIPv1, RIPv2, Static Routing, Route Redistribution and Floating Static Routes

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1. Introduction:

This project demonstrates the implementation of a hybrid enterprise network using multiple routing technologies. The network integrates OSPF Multi-Area Routing, RIP Version 1, RIP Version 2, Static Routing, Route Redistribution, Default Static Routing, and Floating Static Routes. An Autonomous System Boundary Router (ASBR) was configured to exchange routing information between different routing domains, allowing seamless communication throughout the network.

The project also implements route redundancy using floating static routes to provide an alternative path in case of primary route failure.

Tools Used:

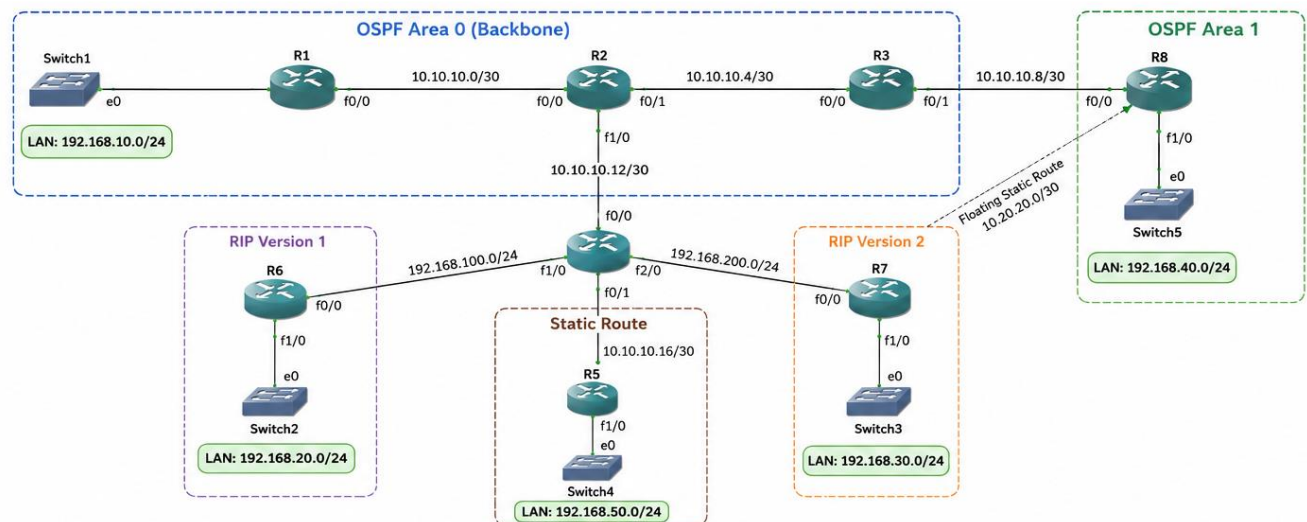
1. GNS3
2. PUTTY

2. Objective

The objectives of this project are:

- ❖ Design a multi-protocol enterprise network.
- ❖ Configure multi-Area OSPF.
- ❖ Configure RIP Version 1.
- ❖ Configure RIP Version 2.
- ❖ Configure Static Routing.
- ❖ Configure an ASBR.
- ❖ Implement Route Redistribution.
- ❖ Configure a Default Static Route.
- ❖ Configure Floating Static Routes.
- ❖ Verify end-to-end connectivity between all LANs.

3. Network Topology:



The network consists of:

- ❖ 8 Cisco routers
- ❖ Multi-Area OSPF network (Area 0 and Area 1)
- ❖ One Area Border Router (R3) & One Autonomous System Boundary Router (R4)
- ❖ One RIPv1 & One RIPv2 routing domain
- ❖ One Static Routing domain & One Floating Static Route for route redundancy
- ❖ One Default Static Route

- ❖ Multiple LAN segments connected to different routers
- ❖ Point-to-point links between routers using /30 subnets
- ❖ Route Redistribution between OSPF, RIP, and Static Routing

4. IP Addressing Plan:

A structured IP addressing scheme was implemented to ensure proper communication between all routers and LANs. The assigned IP addresses were used to configure routing protocols and establish connectivity throughout the network.

Key Points:

- Unique IP addresses were assigned to all router interfaces.
- /24 subnet mask was used for LAN networks.
- /30 subnet mask was used for point-to-point links.
- The addressing scheme prevented IP conflicts and ensured efficient communication.

IP ADDRESSING PLAN				
Device / Link	Interface	IP Address	Subnet Mask	Network
R1 – Switch1 (LAN)	f0/0	192.168.10.1	255.255.255.0	192.168.10.0/24
R1 – R2	f0/0	10.10.10.1	255.255.255.252	10.10.10.0/30
R2 – R1	f0/1	10.10.10.2	255.255.255.252	10.10.10.0/30
R3 – R2	f0/0	10.10.10.5	255.255.255.252	10.10.10.4/30
R3 – R8	f0/0	10.10.10.6	255.255.255.252	10.10.10.8/30
R8 – R3	f0/0	10.10.10.10	255.255.255.252	10.10.10.8/30
R2 – R4	f1/0	10.10.10.13	255.255.255.252	10.10.10.12/30
R4 – R2	f0/0	10.10.10.14	255.255.255.252	10.10.10.12/30
R4 – R5	f0/1	10.10.10.17	255.255.255.252	10.10.10.16/30
R5 – R4	f0/0	10.10.10.18	255.255.255.252	10.10.10.16/30
R4 – R6 (RIPv1)	f1/0	192.168.100.1	255.255.255.0	192.168.100.0/24
R6 – R4 (RIPv1)	f0/0	192.168.100.2	255.255.255.0	192.168.100.0/24
R4 – R7 (RIPv2)	f2/0	192.168.200.1	255.255.255.0	192.168.200.0/24
R7 – R4 (RIPv2)	f0/0	192.168.200.2	255.255.255.0	192.168.200.0/24
R6 – Switch2 (LAN)	f1/0	192.168.20.1	255.255.255.0	192.168.20.0/24
R7 – Switch3 (LAN)	f1/0	192.168.30.1	255.255.255.0	192.168.30.0/24
R8 – Switch5 (LAN)	f1/0	192.168.40.1	255.255.255.0	192.168.40.0/24
R5 – Switch4 (LAN)	f1/0	192.168.50.1	255.255.255.0	192.168.50.0/24
Floating Link (Backup)	–	10.20.20.1 (R8)	255.255.255.252	10.20.20.0/30
		10.20.20.2 (R3)	255.255.255.252	

Each router interface was assigned an IP address according to the addressing plan. LAN segments use /24 subnet masks, while point-to-point links use /30 subnet masks to efficiently utilize IP address space.

5. Routing Protocol Allocation

Different routing protocols were assigned to routers based on the network design and project requirements. OSPF was implemented as the primary dynamic routing protocol, while RIP Version 1, RIP Version 2, and Static Routing were configured in separate routing domains. Router R3 was configured as the Area Border Router (ABR), and Router R4 was configured as the Autonomous System Boundary Router (ASBR) to enable route redistribution between different routing protocols.

Figure 3: Routing Protocol Allocation Table:

ROUTING PROTOCOL SUMMARY		
Area / Segment	Routing Protocol	Description
OSPF Area 0 (Backbone)	OSPF (Process 1)	Backbone area connecting R1, R2, R3 and ASBR R4.
OSPF Area 1	OSPF (Process 1)	Connected to Area 0 via R3. Includes LAN 192.168.40.0/24.
RIP Version 1	RIPv1	Configured on R6 with LAN 192.168.20.0/24.
RIP Version 2	RIPv2	Configured on R7 with LAN 192.168.30.0/24.
Static Route Segment	Static Routing	R5 network 192.168.50.0/24 is added using static route.
Route Redistribution (on R4)	• RIP (v1 & v2) → OSPF • Static → OSPF • OSPF → RIP (v1 & v2) • Static → RIP (v1 & v2)	R4 acts as ASBR and redistributes routes between domains.
Redundancy / Backup	Floating Static Route	Backup link between R3 and R8 (10.20.20.0/30) configured with floating static route.

Explanation

The routing protocol allocation table shows the routing protocol assigned to each router. This design separates the network into different routing domains while allowing communication between them through the ASBR using route redistribution. The use of an ABR also enables communication between OSPF Area 0 and Area 1.

6. OSPF Configuration

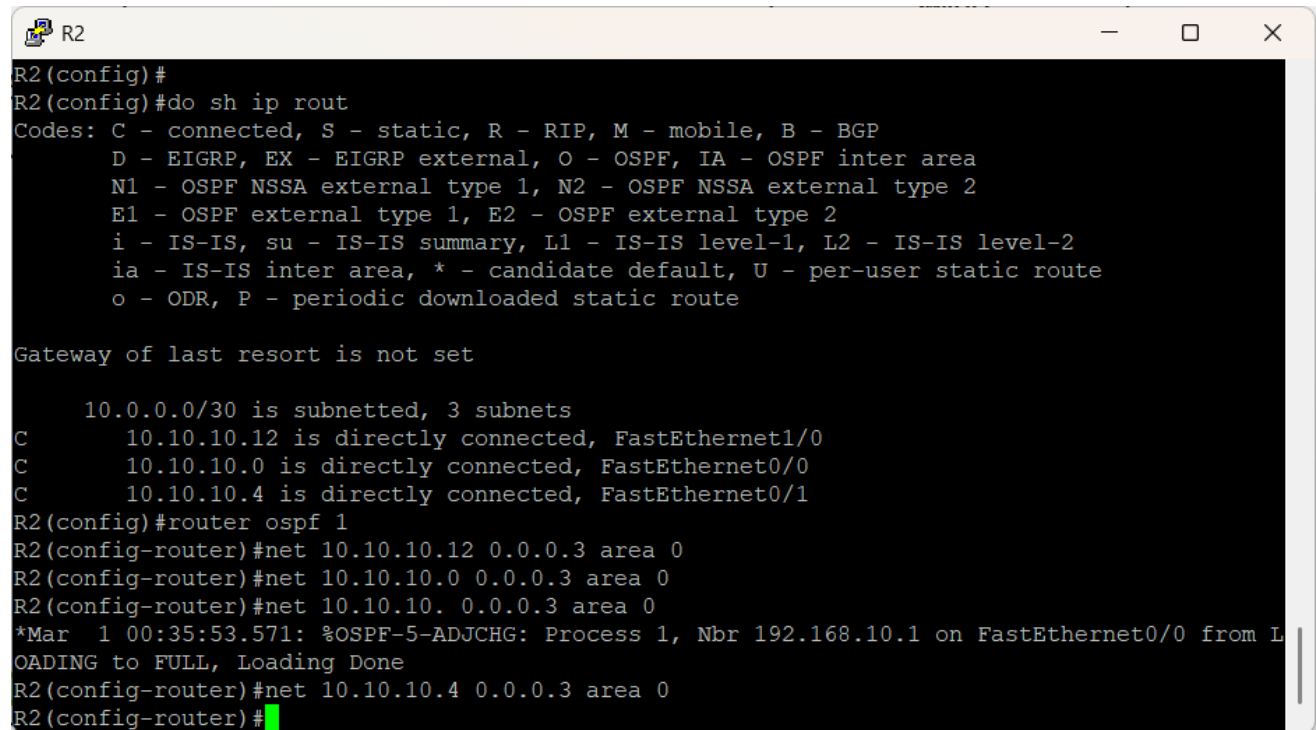
6.1 OSPF Backbone Area

Routers R1 and R2 were configured as members of OSPF Area 0, which serves as the backbone area of the network. Appropriate network statements were configured to advertise directly connected interfaces. After configuration, OSPF neighbor relationships were successfully established, enabling routing information to be exchanged across the backbone.

Key Points

- OSPF Process ID 1 was configured.
- Both routers were assigned to Area 0.
- Point-to-point links were advertised.
- OSPF neighbor relationship was established successfully.
- Routes were exchanged through the backbone area.

Figure 4: Backbone Area Routers (R1, R2) Configuration

A screenshot of a terminal window titled 'R2' showing the configuration of OSPF Area 0 on Router R2. The terminal output includes the command 'do sh ip rout' which displays the IP routing table for the 10.0.0.0/30 network, showing three subnets: 10.10.10.12, 10.10.10.0, and 10.10.10.4. The configuration then shows the OSPF process being started with 'router ospf 1', and three network statements being added: 'net 10.10.10.12 0.0.0.3 area 0', 'net 10.10.10.0 0.0.0.3 area 0', and 'net 10.10.10. 0.0.0.3 area 0'. A message indicates that the OSPF process is loading to FULL. The terminal ends with the prompt 'R2(config-router)#' and a green cursor.

```
R2
R2(config)#
R2(config)#do sh ip rout
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 3 subnets
C       10.10.10.12 is directly connected, FastEthernet1/0
C       10.10.10.0 is directly connected, FastEthernet0/0
C       10.10.10.4 is directly connected, FastEthernet0/1
R2(config)#router ospf 1
R2(config-router)#net 10.10.10.12 0.0.0.3 area 0
R2(config-router)#net 10.10.10.0 0.0.0.3 area 0
R2(config-router)#net 10.10.10. 0.0.0.3 area 0
*Mar  1 00:35:53.571: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on FastEthernet0/0 from L
LOADING to FULL, Loading Done
R2(config-router)#net 10.10.10.4 0.0.0.3 area 0
R2(config-router)#
```

The screenshot shows the successful OSPF Area 0 configuration on Router R2.

6.2 Area Border Router (ABR)

Router R3 was configured as an Area Border Router (ABR). It connects OSPF Area 0 and Area 1 by advertising networks into their respective areas. This allows routing information to be exchanged between both OSPF areas while maintaining a hierarchical network design.

Key Points

- R3 functions as the Area Border Router.
- Connects Area 0 and Area 1.
- Advertises networks in both areas.
- Enables inter-area communication.

Figure 5: OSPF ABR Configuration on R3



```
R3
R3(config-if)#
R3(config-if)#ex
R3(config)#
R3(config)#do sh ip rout
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 2 subnets
C       10.10.10.8 is directly connected, FastEthernet0/1
C       10.10.10.4 is directly connected, FastEthernet0/0
R3(config)#router ospf 1
R3(config-router)#net 10.10.10.4 0.0.0.3 a 0
R3(config-router)#net 10.10.10. 0.0.0.3 a 1
*Mar  1 00:36:12.875: %OSPF-5-ADJCHG: Process 1, Nbr 10.10.10.13 on FastEthernet
0/0 from LOADING to FULL, Loading Done
R3(config-router)#net 10.10.10.8 0.0.0.3 a 1
R3(config-router)#
```

The screenshot illustrates the OSPF configuration on Router R3. The router advertises interfaces in both Area 0 and Area 1, allowing it to exchange routing information.

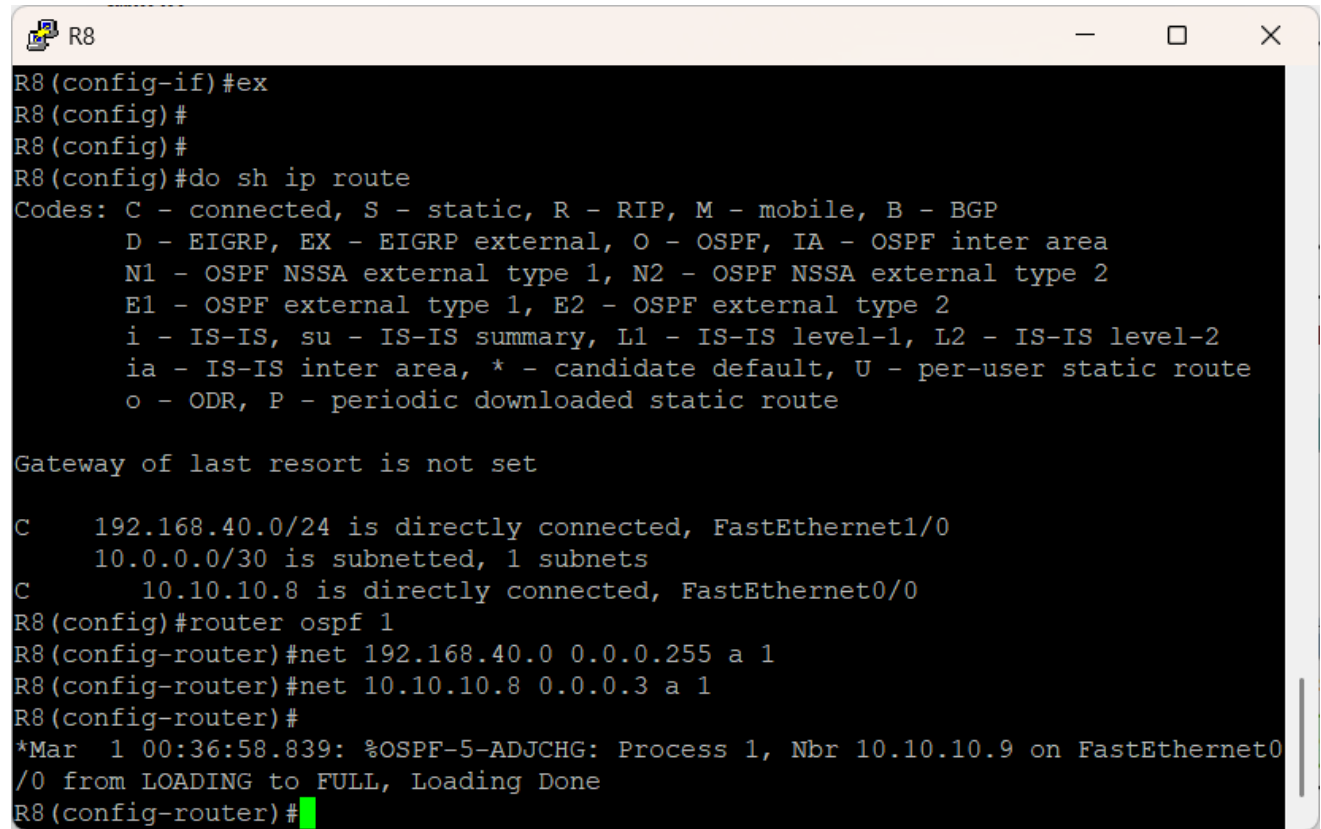
6.3 OSPF Area 1

Router R8 was configured within OSPF Area 1. The router advertises its directly connected LAN network and establishes adjacency with the Area Border Router. This enables communication between Area 1 and the backbone network.

Key Points

- R8 belongs to OSPF Area 1.
- Advertises its directly connected LAN.
- Forms adjacency with the ABR.
- Learns routes from other OSPF areas.

Figure 6: OSPF Area 1 Configuration on R8



```
R8
R8(config-if)#ex
R8(config)#
R8(config)#
R8(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.40.0/24 is directly connected, FastEthernet1/0
     10.0.0.0/30 is subnetted, 1 subnets
C      10.10.10.8 is directly connected, FastEthernet0/0
R8(config)#router ospf 1
R8(config-router)#net 192.168.40.0 0.0.0.255 a 1
R8(config-router)#net 10.10.10.8 0.0.0.3 a 1
R8(config-router)#
*Mar  1 00:36:58.839: %OSPF-5-ADJCHG: Process 1, Nbr 10.10.10.9 on FastEthernet0
/0 from LOADING to FULL, Loading Done
R8(config-router)#
```

The screenshot shows the successful configuration of OSPF Area 1 on Router R8. After the configuration, Router R8 exchanged routing information with the ABR, providing connectivity between Area 1 and the backbone network.

6.4 OSPF Verification

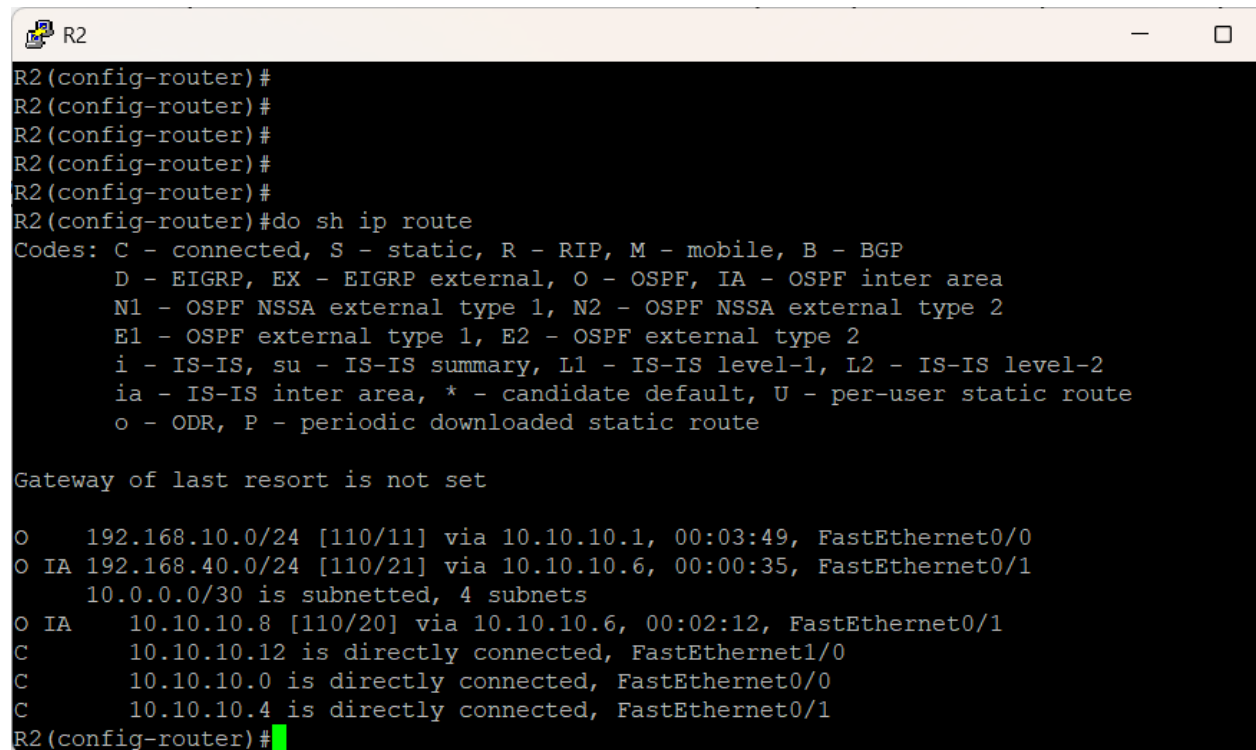
Description

The OSPF routing table was verified to ensure that routing information was successfully exchanged between Area 0 and Area 1. The presence of inter-area routes confirms that the ABR is functioning correctly.

Key Points

- Routing table verified successfully.
- Inter-area routes appeared as O IA.
- Communication between Area 0 and Area 1 was confirmed.
- OSPF routing operated as expected.

Figure 7: Routing Table Verification on R2



```
R2
R2(config-router)#
R2(config-router)#
R2(config-router)#
R2(config-router)#
R2(config-router)#
R2(config-router)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.10.0/24 [110/11] via 10.10.10.1, 00:03:49, FastEthernet0/0
O IA 192.168.40.0/24 [110/21] via 10.10.10.6, 00:00:35, FastEthernet0/1
      10.0.0.0/30 is subnetted, 4 subnets
O IA   10.10.10.8 [110/20] via 10.10.10.6, 00:02:12, FastEthernet0/1
C      10.10.10.12 is directly connected, FastEthernet1/0
C      10.10.10.0 is directly connected, FastEthernet0/0
C      10.10.10.4 is directly connected, FastEthernet0/1
R2(config-router)#
```

Explanation: The routing table of Router R2 displays O IA (Inter-Area) routes, confirming that networks from Area 1 were successfully learned through the Area Border Router. This verifies the correct implementation of multi-Area OSPF.

7. RIP Configuration

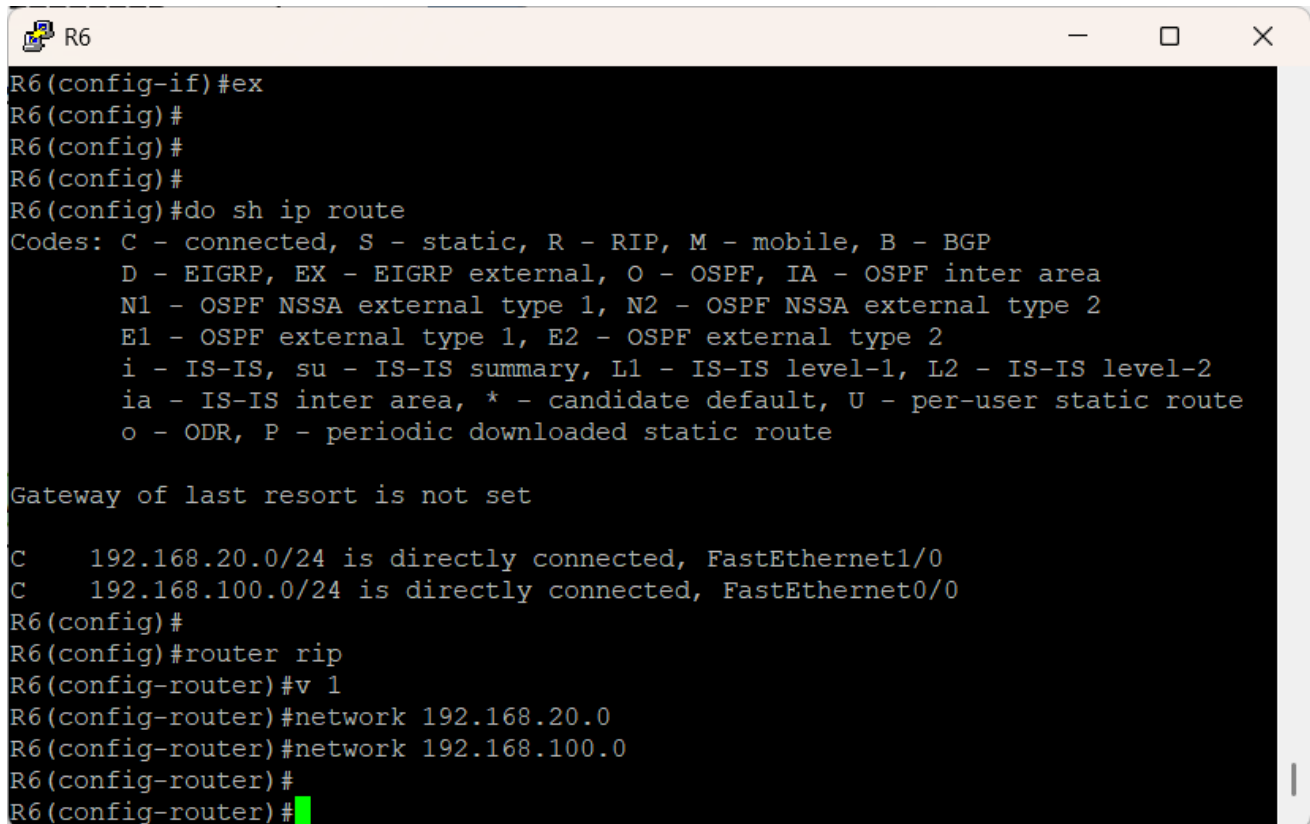
Routing Information Protocol (RIP) was implemented to provide dynamic routing within separate routing domains. Router R6 was configured with RIP Version 1, while Router R7 was configured with RIP Version 2. Both routing domains were connected to the Autonomous System Boundary Router (ASBR), allowing route exchange with OSPF and Static Routing through route redistribution.

7.1 RIP Version 1 Configuration

Key Points

- RIP Version 1 was configured on Router R6.
- Directly connected networks were advertised.
- Classful routing was used.
- Routes were exchanged with the ASBR.

Figure 8: RIP Version 1 Configuration on R6



```
R6
R6(config-if)#ex
R6(config)#
R6(config)#
R6(config)#
R6(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.20.0/24 is directly connected, FastEthernet1/0
C    192.168.100.0/24 is directly connected, FastEthernet0/0
R6(config)#
R6(config)#router rip
R6(config-router)#v 1
R6(config-router)#network 192.168.20.0
R6(config-router)#network 192.168.100.0
R6(config-router)#
R6(config-router)#
```

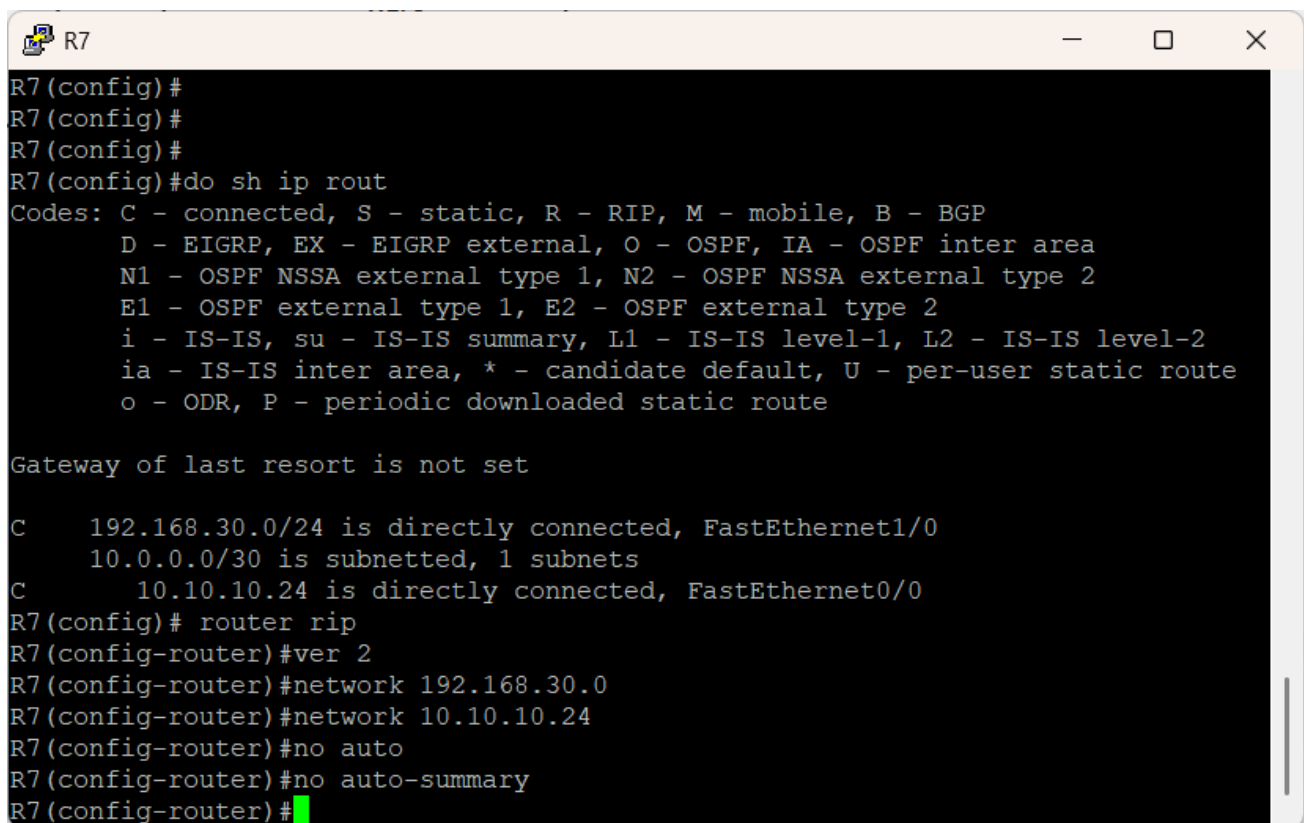
The screenshot shows the successful configuration of RIPv1 on R6. The router advertises its directly connected networks and exchanges routing information with the ASBR.

7.2 RIP Version 2 Configuration

Key Points

- RIP Version 2 was configured on Router R7.
- Directly connected networks were advertised.
- no auto-summary was configured.
- Routes were exchanged with the ASBR.

Figure 9: RIP Version 2 Configuration on R7



```
R7
R7(config)#
R7(config)#
R7(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.30.0/24 is directly connected, FastEthernet1/0
     10.0.0.0/30 is subnetted, 1 subnets
C      10.10.10.24 is directly connected, FastEthernet0/0
R7(config)# router rip
R7(config-router)#ver 2
R7(config-router)#network 192.168.30.0
R7(config-router)#network 10.10.10.24
R7(config-router)#no auto
R7(config-router)#no auto-summary
R7(config-router)#
```

The screenshot shows the successful configuration of RIP Version 2 on Router R7. RIP Version 2 supports classless routing and subnet mask information, enabling accurate route advertisement. The router exchanges routing information with the ASBR for communication with other routing domains.

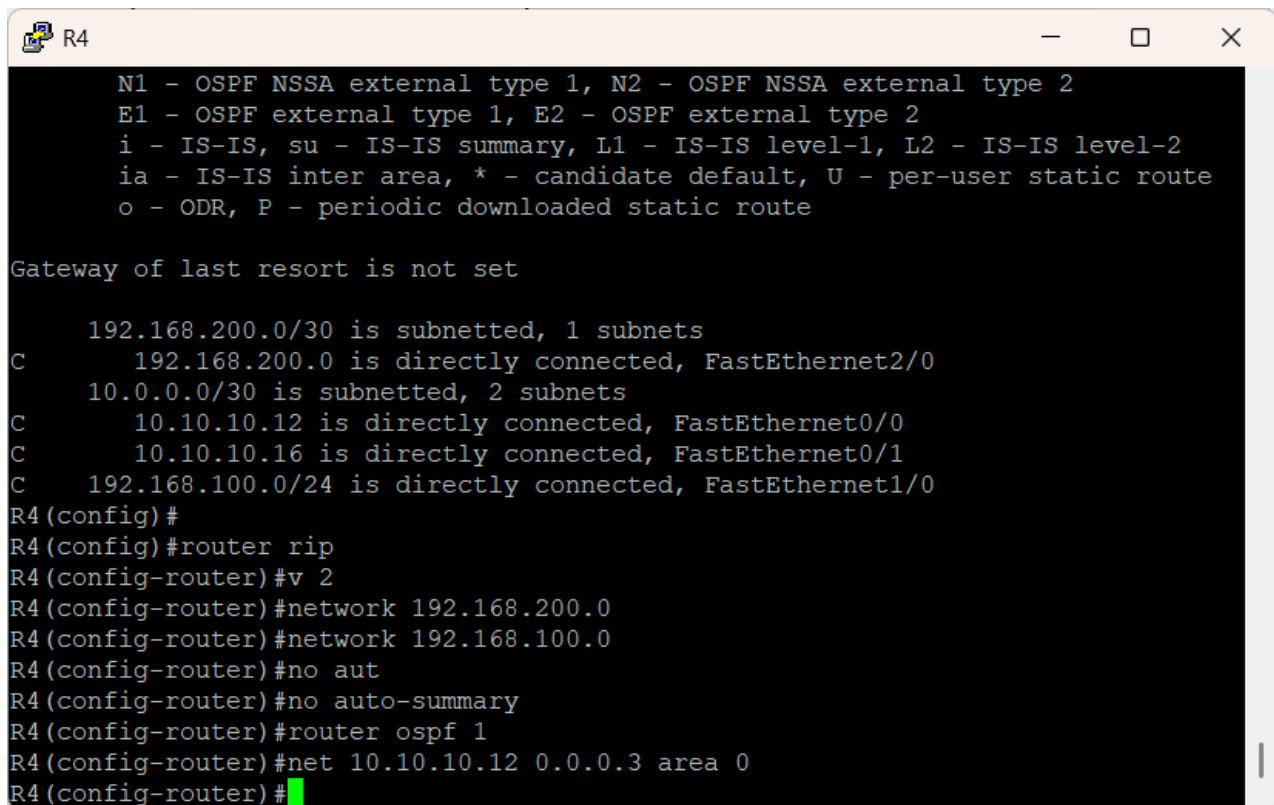
8. ASBR Configuration

Router R4 was configured as the Autonomous System Boundary Router (ASBR) to connect different routing domains within the network. It serves as the central router between OSPF, RIP, and Static Routing. In addition to running multiple routing protocols, the ASBR performs route redistribution, allowing routers using different protocols to exchange routing information and communicate successfully.

Key Points

- Router R4 was configured as the ASBR.
- OSPF and RIP were configured on the same router.
- Connected the OSPF, RIP, and Static Routing domains.
- Enabled communication between different routing protocols.
- Performed route redistribution between routing domains.

Figure 10: ASBR Configuration on Router R4

A screenshot of a terminal window titled 'R4' showing the configuration of Router R4. The terminal displays a legend for configuration codes: N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2, E1 - OSPF external type 1, E2 - OSPF external type 2, i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area, * - candidate default, U - per-user static route, o - ODR, P - periodic downloaded static route. It then shows the status 'Gateway of last resort is not set' and a list of connected networks: 192.168.200.0/30 (1 subnet), 10.0.0.0/30 (2 subnets), and 192.168.100.0/24. The configuration commands for RIP and OSPF are shown, including enabling both protocols, setting version 2 for RIP, and configuring network statements for both protocols. The terminal ends with the prompt 'R4(config-router)#' and a green cursor.

```
R4
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  192.168.200.0/30 is subnetted, 1 subnets
C       192.168.200.0 is directly connected, FastEthernet2/0
  10.0.0.0/30 is subnetted, 2 subnets
C       10.10.10.12 is directly connected, FastEthernet0/0
C       10.10.10.16 is directly connected, FastEthernet0/1
C       192.168.100.0/24 is directly connected, FastEthernet1/0
R4(config)#
R4(config)#router rip
R4(config-router)#v 2
R4(config-router)#network 192.168.200.0
R4(config-router)#network 192.168.100.0
R4(config-router)#no aut
R4(config-router)#no auto-summary
R4(config-router)#router ospf 1
R4(config-router)#net 10.10.10.12 0.0.0.3 area 0
R4(config-router)#
```

The screenshot shows the routing protocol configuration on Router R4. The router was configured with both OSPF and RIP, enabling it to connect multiple routing domains.

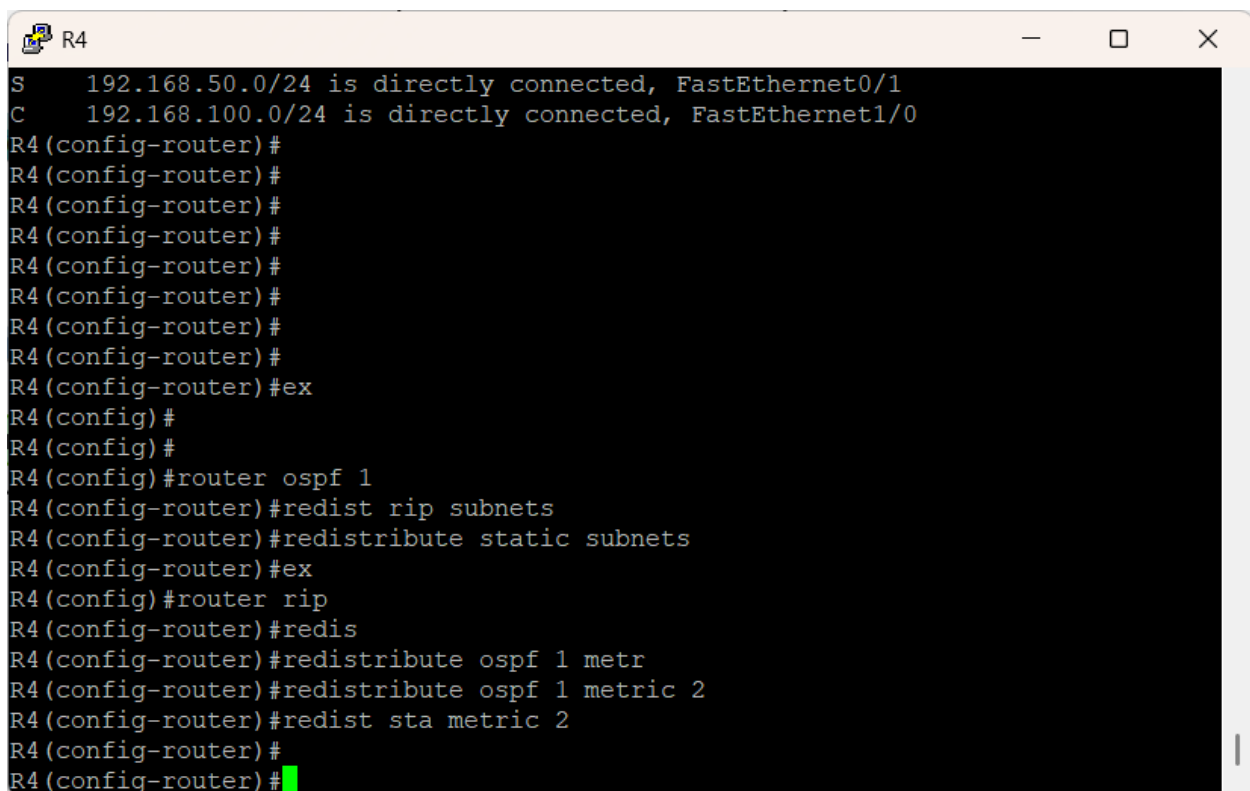
9. Route Redistribution

Route redistribution was configured on Router R4 (ASBR) to enable communication between different routing protocols. This allowed all routing domains to exchange routing information and achieve end-to-end connectivity.

Key Points

- RIP routes were redistributed into OSPF.
- Static routes were redistributed into OSPF.
- OSPF routes were redistributed into RIP.
- Static routes were redistributed into RIP.
- Route metrics were configured where required.
- Full communication was established between different routing domains.

Figure 11: Route Redistribution Configuration on R4

A screenshot of a terminal window titled 'R4' showing the configuration of route redistribution on a router. The terminal output includes status messages for directly connected networks, followed by a series of configuration commands in configuration mode. The commands configure OSPF, redistribute static and RIP routes into OSPF with specific metrics, and then configure RIP to redistribute OSPF and static routes. The terminal ends with a green cursor on the 'R4(config-router)#' prompt.

```
S    192.168.50.0/24 is directly connected, FastEthernet0/1
C    192.168.100.0/24 is directly connected, FastEthernet1/0
R4(config-router)#
R4(config-router)#
R4(config-router)#
R4(config-router)#
R4(config-router)#
R4(config-router)#
R4(config-router)#
R4(config-router)#ex
R4(config)#
R4(config)#
R4(config)#router ospf 1
R4(config-router)#redist rip subnets
R4(config-router)#redistribute static subnets
R4(config-router)#ex
R4(config)#router rip
R4(config-router)#redis
R4(config-router)#redistribute ospf 1 metr
R4(config-router)#redistribute ospf 1 metric 2
R4(config-router)#redist sta metric 2
R4(config-router)#
R4(config-router)#
```

The screenshot shows the redistribution configuration on the ASBR. By redistributing routes between OSPF, RIP, and Static Routing, routers running different routing protocols were able to learn remote networks and communicate with each other. This configuration successfully integrated all routing domains into a single interconnected network.

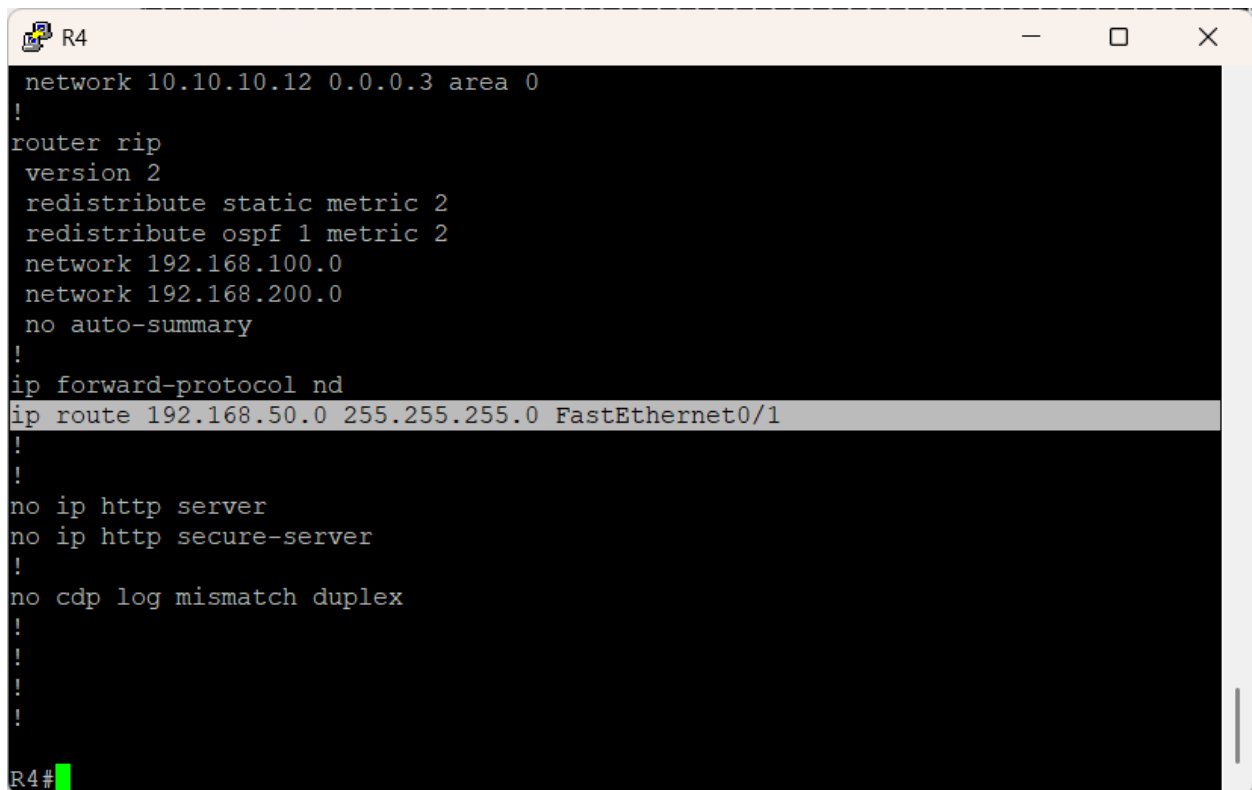
10. Static Routing Configuration

Static routing was implemented on Router R4 to provide connectivity with the rest of the network. Since R4 was not running any dynamic routing protocol, a default static route was configured to forward all unknown traffic to Router R4 (ASBR). This allowed the static routing domain to communicate with OSPF and RIP networks through the ASBR.

Key Points

- Static routing was configured on Router R4.
- A default static route was configured.
- Unknown traffic was forwarded to Router R4 (ASBR).
- Communication with dynamic routing domains was successfully established.
- Connectivity was verified using end-to-end ping tests.

Figure 12: Static Route Configuration on Router R4



```
network 10.10.10.12 0.0.0.3 area 0
!
router rip
version 2
redistribute static metric 2
redistribute ospf 1 metric 2
network 192.168.100.0
network 192.168.200.0
no auto-summary
!
ip forward-protocol nd
ip route 192.168.50.0 255.255.255.0 FastEthernet0/1
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
R4#
```

Default Static Route Configuration:

```
Port-channel      Ethernet Channel of interfaces
Tunnel            Tunnel interface
Vif               PGM Multicast Host interface
Virtual-PPP       Virtual PPP interface
Virtual-TokenRing Virtual TokenRing
Vlan              Vlan IEEE 802.1q
XTagATM           Extended Tag ATM interface
vmi              Virtual Multipoint Interface

R5(config)#ip route 0.0.0.0 0.0.0.0
R5(config)#ip route 0.0.0.0 0.0.0.0 f0/0
R5(config)#
R5(config)#
R5(config)#          FOR DEFAULT ROUTE
```

The screenshot shows the default static route configured on Router R4. By forwarding unknown destinations to the ASBR, Router R5 was able to communicate with networks learned through OSPF and RIP. This simplified the configuration while maintaining full network connectivity.

11. Connectivity Verification:

```
R8#
R8#
R8#
R8#
R8#
R8#ping 192.168.50.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 104/119/128 ms
R8#
R8#
R8#
R8#          PING SUCCESSFUL AFTER DEFAULT ROUTE
```

After configuring the default static route, a successful ping was performed from Router R8 to the LAN connected to Router R5. The successful ICMP replies confirm that the default route is functioning correctly and that communication between the OSPF and Static Routing domains has been established successfully.

12. Floating Static Route:

A floating static route was configured to provide backup connectivity between the RIPv2 domain and OSPF Area 1.

Key Points

- A backup link was established between the RIPv2 domain and OSPF Area 1.
- A floating static route was configured on Router R8.
- Administrative Distance (AD) 200 was assigned to the backup route.
- The primary route is preferred during normal operation.
- The backup route becomes active only when the primary route fails.
- Route redundancy was successfully implemented.

Figure 14: Floating Static Route Topology

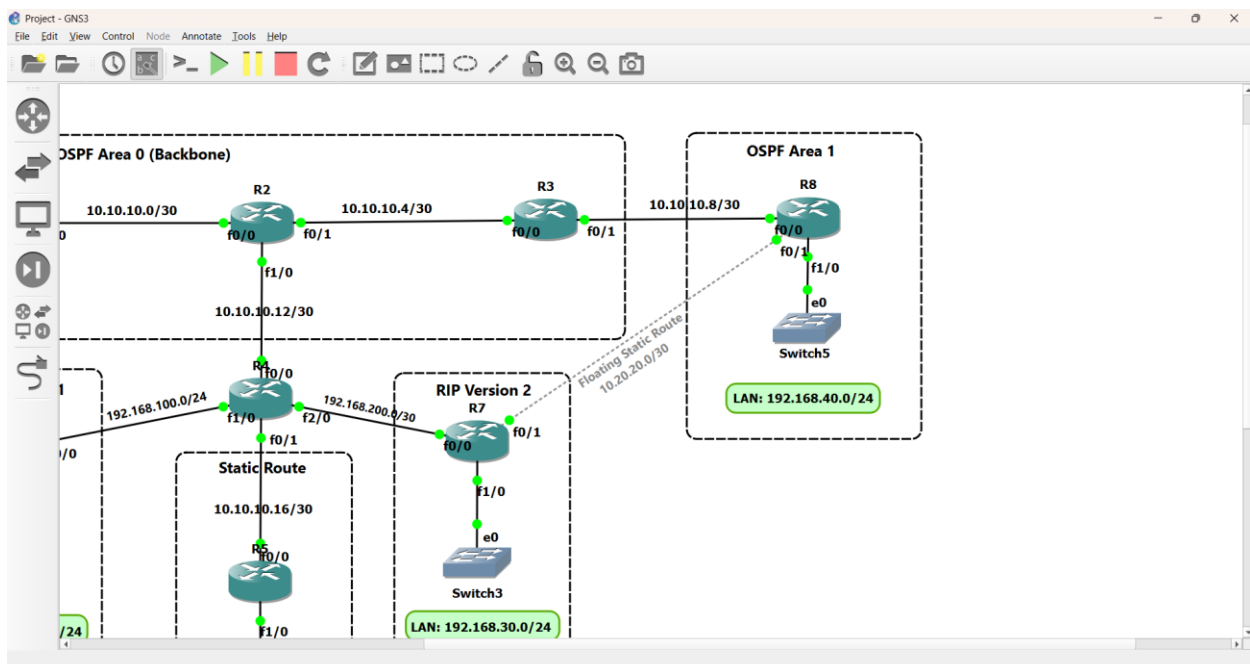
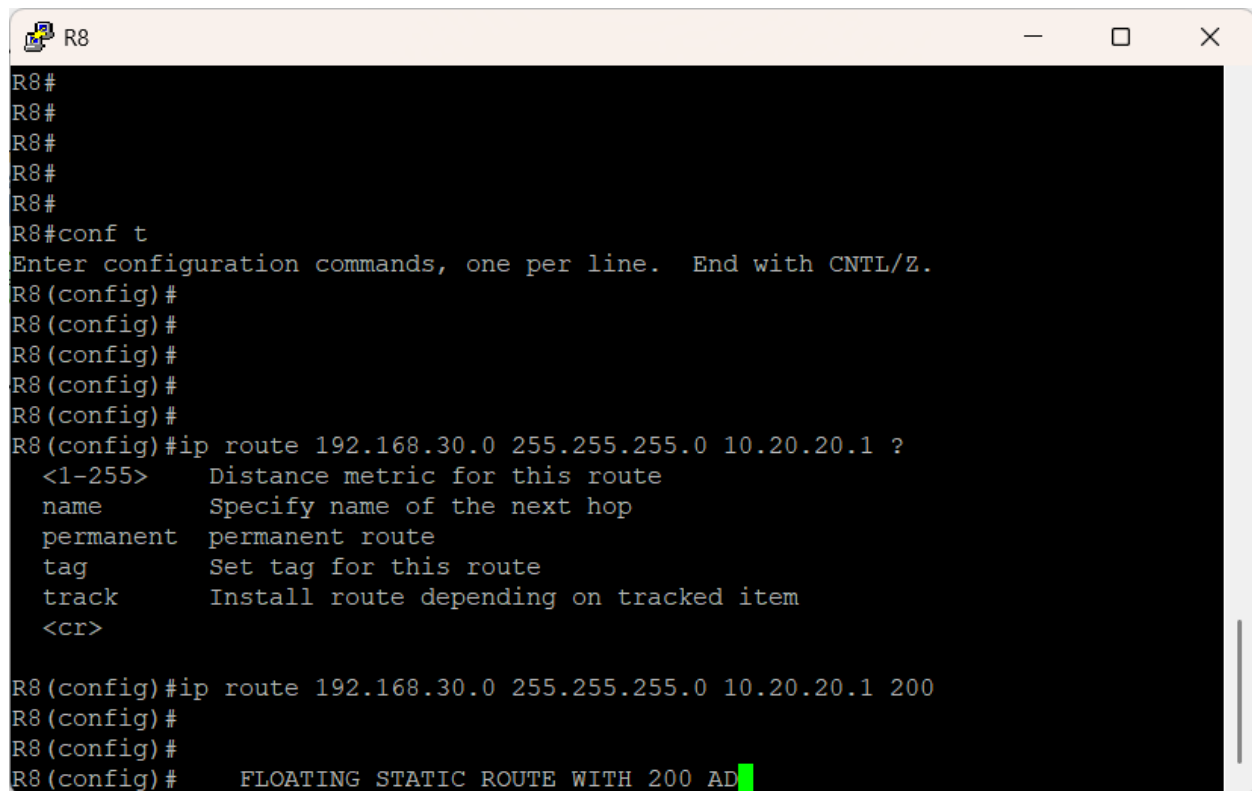


Figure 15: Floating Static Route Configuration on R8



```
R8#
R8#
R8#
R8#
R8#
R8#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R8(config)#
R8(config)#
R8(config)#
R8(config)#
R8(config)#
R8(config)#ip route 192.168.30.0 255.255.255.0 10.20.20.1 ?
    <1-255>  Distance metric for this route
    name      Specify name of the next hop
    permanent permanent route
    tag       Set tag for this route
    track     Install route depending on tracked item
    <cr>

R8(config)#ip route 192.168.30.0 255.255.255.0 10.20.20.1 200
R8(config)#
R8(config)#
R8(config)#      FLOATING STATIC ROUTE WITH 200 AD
```

Explanation

The screenshots show the backup topology and the floating static route configured on Router R8. The route was assigned an Administrative Distance of 200, ensuring it remains inactive while the primary route is available. In the event of a primary path failure, the floating static route provides an alternative path, improving the reliability and availability of the network.

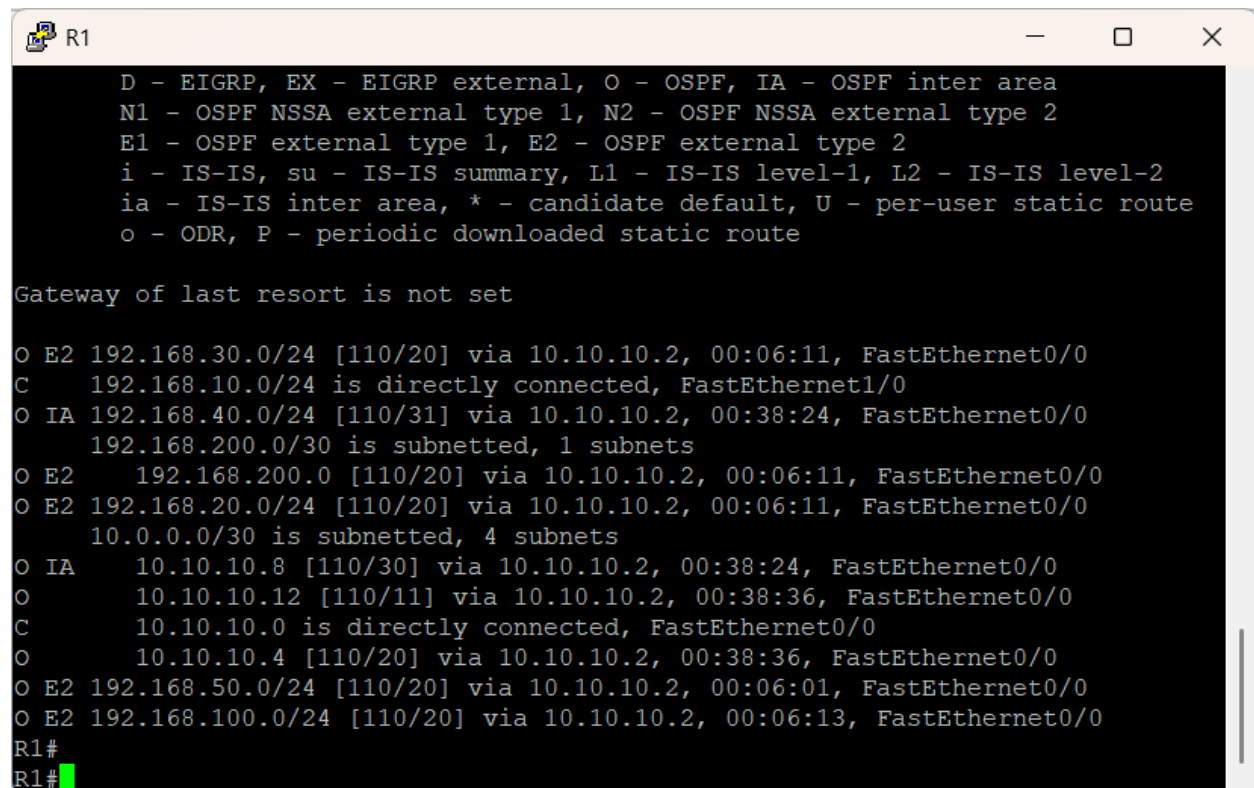
13. Final Verification

After completing all routing configurations, verification was performed to ensure that the network was operating correctly. Routing tables and connectivity tests confirmed that routes were successfully exchanged between OSPF, RIP, and Static Routing domains. End-to-end communication was verified across all LAN networks.

Key Points

- Routing table verified on Router R1 & R8.
- Primary and backup routes were confirmed.
- End-to-end connectivity was successfully tested.
- All LAN networks were reachable.
- The network operated as expected after route redistribution and backup route configuration.

Figure 16: Routing Table of R1

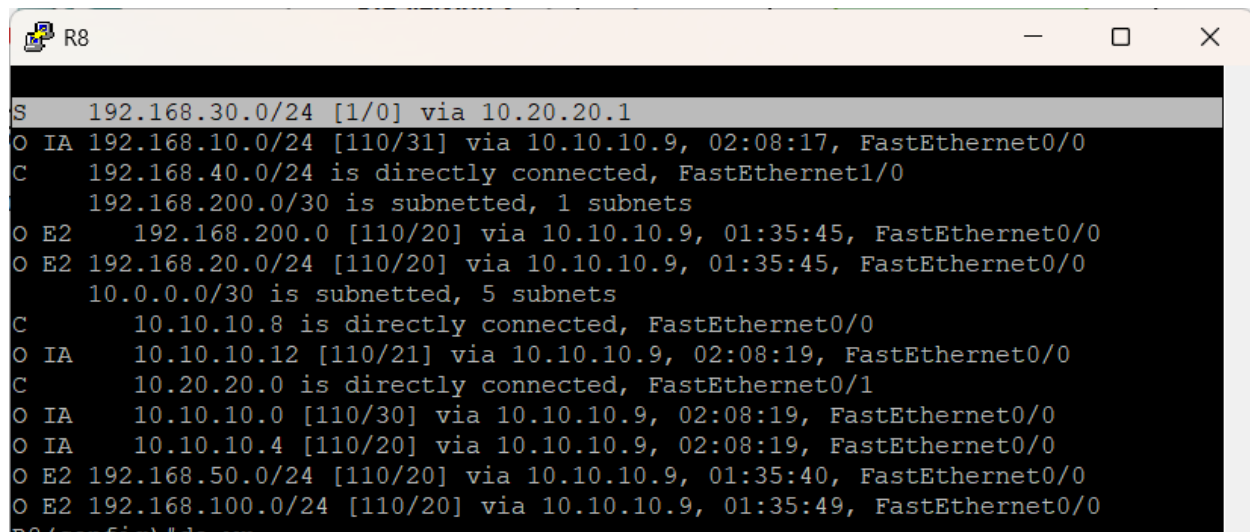


```
R1
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O E2 192.168.30.0/24 [110/20] via 10.10.10.2, 00:06:11, FastEthernet0/0
C    192.168.10.0/24 is directly connected, FastEthernet1/0
O IA 192.168.40.0/24 [110/31] via 10.10.10.2, 00:38:24, FastEthernet0/0
    192.168.200.0/30 is subnetted, 1 subnets
O E2   192.168.200.0 [110/20] via 10.10.10.2, 00:06:11, FastEthernet0/0
O E2 192.168.20.0/24 [110/20] via 10.10.10.2, 00:06:11, FastEthernet0/0
    10.0.0.0/30 is subnetted, 4 subnets
O IA   10.10.10.8 [110/30] via 10.10.10.2, 00:38:24, FastEthernet0/0
O     10.10.10.12 [110/11] via 10.10.10.2, 00:38:36, FastEthernet0/0
C     10.10.10.0 is directly connected, FastEthernet0/0
O     10.10.10.4 [110/20] via 10.10.10.2, 00:38:36, FastEthernet0/0
O E2 192.168.50.0/24 [110/20] via 10.10.10.2, 00:06:01, FastEthernet0/0
O E2 192.168.100.0/24 [110/20] via 10.10.10.2, 00:06:13, FastEthernet0/0
R1#
R1#
```

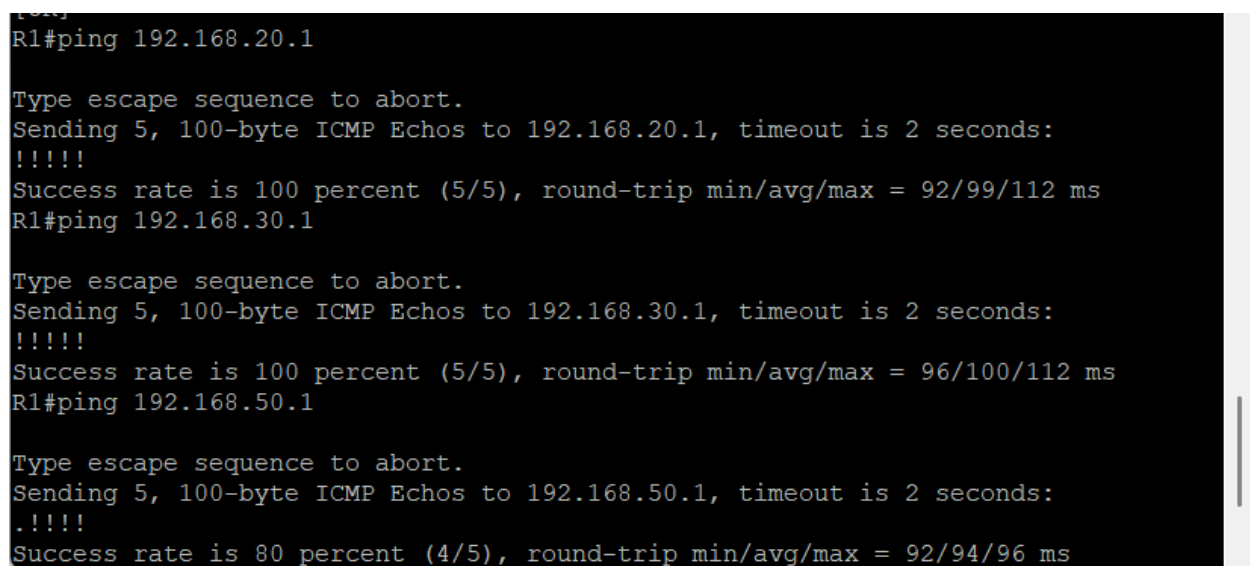
Figure 17: Routing Table of R8



```
R8
S 192.168.30.0/24 [1/0] via 10.20.20.1
O IA 192.168.10.0/24 [110/31] via 10.10.10.9, 02:08:17, FastEthernet0/0
C 192.168.40.0/24 is directly connected, FastEthernet1/0
192.168.200.0/30 is subnetted, 1 subnets
O E2 192.168.200.0 [110/20] via 10.10.10.9, 01:35:45, FastEthernet0/0
O E2 192.168.20.0/24 [110/20] via 10.10.10.9, 01:35:45, FastEthernet0/0
10.0.0.0/30 is subnetted, 5 subnets
C 10.10.10.8 is directly connected, FastEthernet0/0
O IA 10.10.10.12 [110/21] via 10.10.10.9, 02:08:19, FastEthernet0/0
C 10.20.20.0 is directly connected, FastEthernet0/1
O IA 10.10.10.0 [110/30] via 10.10.10.9, 02:08:19, FastEthernet0/0
O IA 10.10.10.4 [110/20] via 10.10.10.9, 02:08:19, FastEthernet0/0
O E2 192.168.50.0/24 [110/20] via 10.10.10.9, 01:35:40, FastEthernet0/0
O E2 192.168.100.0/24 [110/20] via 10.10.10.9, 01:35:49, FastEthernet0/0
R8(config)#do sh
```

The routing table on Router R8 confirms that the primary routes are preferred under normal conditions, while the floating static route remains available as a backup.

Figure 18: End-to-End Ping Verification



```
R1#ping 192.168.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/99/112 ms
R1#ping 192.168.30.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 96/100/112 ms
R1#ping 192.168.50.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.1, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 92/94/96 ms
```

Successful ping results from Router R1 to all LAN networks verify that OSPF, RIP Version 1, RIP Version 2, Static Routing, Route Redistribution, and Floating Static Routing were configured correctly, providing complete end-to-end network connectivity.

14. Conclusion

This project successfully implemented a hybrid routing environment consisting of OSPF Multi-Area Routing, RIP Version 1, RIP Version 2, Static Routing, Route Redistribution, Default Static Routing, and Floating Static Routes. Router R3 functioned as the Area Border Router, while Router R4 served as the Autonomous System Boundary Router, enabling communication between multiple routing domains.

Route redistribution allowed all routing protocols to exchange network information successfully, while the default static route enabled communication with the static routing domain. The floating static route provided backup connectivity, ensuring network resilience during primary path failure.

The final verification through routing tables and end-to-end ping tests confirmed successful communication between all LAN networks and demonstrated the correct operation of the complete enterprise routing infrastructure.

**** Report Ends Here ****